

Regional Antibiotic Prescribing Patterns in England: Clustering & Explainable AI Analysis

This exploratory analysis investigates regional variation in antibiotic prescribing across 106 UK Sub-ICBs (2024-2025 data), combining unsupervised learning with explainable AI to identify distinct regional "prescribing types" and quantify their socio-economic and clinical determinants.

Exploratory observations

Four prescribing types:

Cluster 0: High Demand, Well-Managed (26 regions)

Cluster 1: Low Intensity, Inefficient (24 regions)

Cluster 2: Low Intensity, Targeted (29 regions)

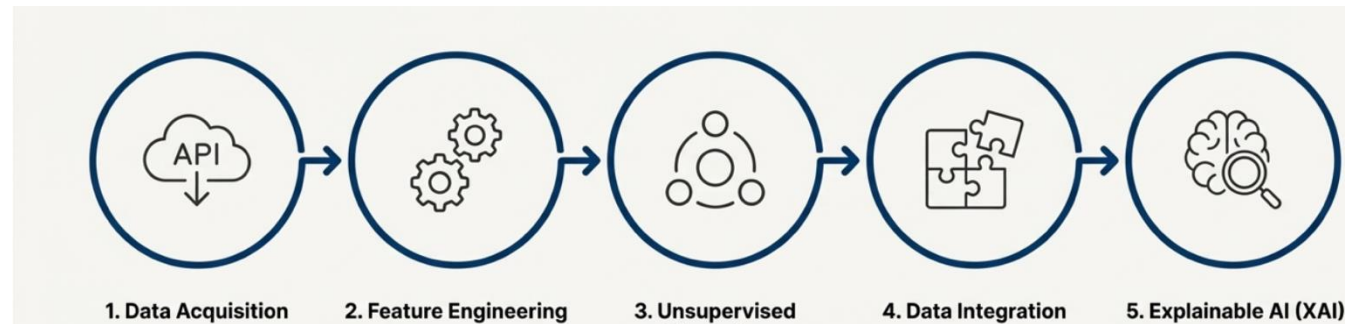
Cluster 3: The Double Burden (27 regions)

Potential future research direction:

time series surveillance and analysis at practice level.

Health economic modelling for budget forecasting

Transferable to other health systems antimicrobial stewardship and health service research



Clustering Pipeline: Identifying Regional Prescribing Patterns

1.Data collection

extracted the latest (2024/25) antibiotic prescribing data for 106 UK Sub-ICBs from the Fingertips API.

2.Features engineering

- Intensity: items per 1000 registered patients.
- Quality: proportion of broad-spectrum antibiotics.
- Seasonal sensitivity: quantified as the ratio of winter prescribing volume to summer
- Trend: quantified by volume yearly percentage change

3.Unsupervised Learning (clustering)

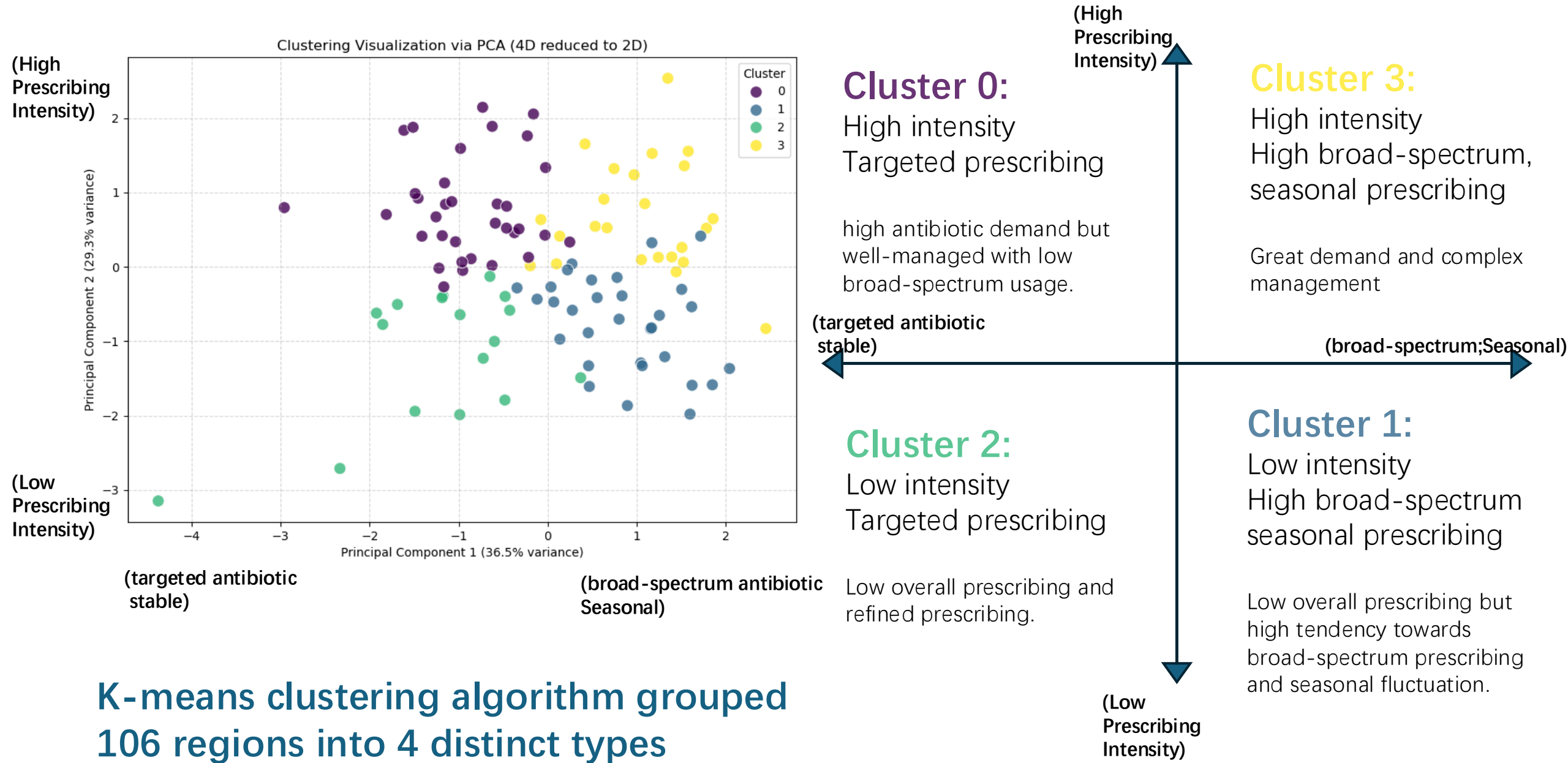
- Algorithm: applied K-Means unsupervised learning to 4-dimensional feature data. The K-means clusters regions based on their prescription behaviors.
- Outcome: 106 regions were grouped into 4 prescribing types.

4.dimensionality reduction & visualization

Used Principal component analysis(PCA) to visualize the 4D cluster in 2D panel.

Interpretation of 2 principal axes.

- Vertical axis (PC2): Prescribing intensity. Represents the amount of antibiotic use. Upward: high intensity; downward: low intensity.
- Horizontal axis (PC1): Prescribing Quality & Seasonality. To the right represents “complex structure” (more broad-spectrum antibiotic, large seasonal fluctuations), while to the left represents more refined use of narrow-spectrum antibiotic and seasonal stability.



Explainable AI pipeline: analysis Socio-economic determinants of general and cluster-specific prescribing

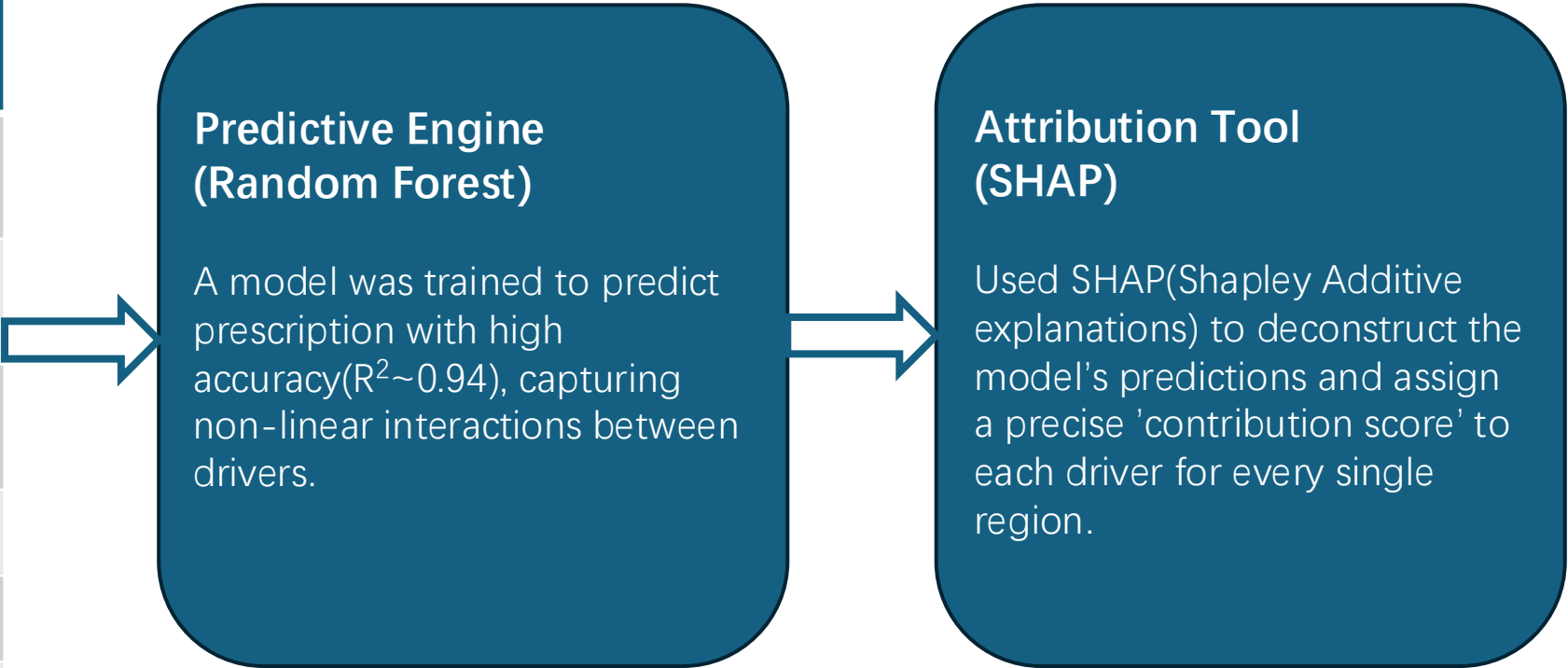
1.Data integration:

Merge prescription data with social determinants:
GP density, IMD, COPD prevalence, and
demographic (aging population & ethnicity)

2. Explainable AI

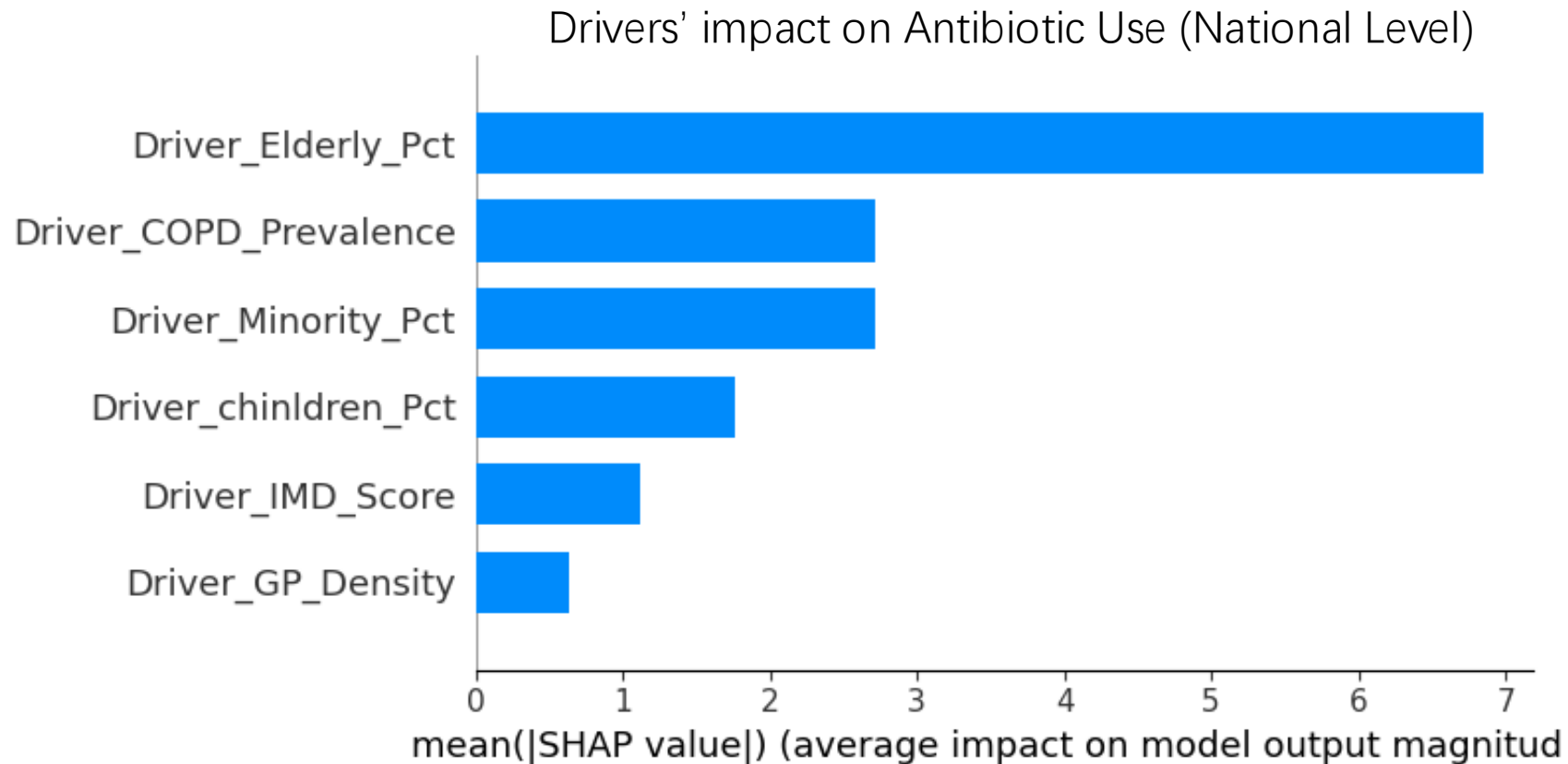
Train a random forest model and use
SHAP to perform determinants'
contribution for each cluster.

Determinants	Data Source
GP workforce density	NHS digital
>65 patients %	NHS digital
<14 patients %	NHS digital
IMD Score	Fingertips
COPD Prevalence	Fingertips
Minority ethnicity %	Nomis



At national level, the elderly population is the most dominant driver

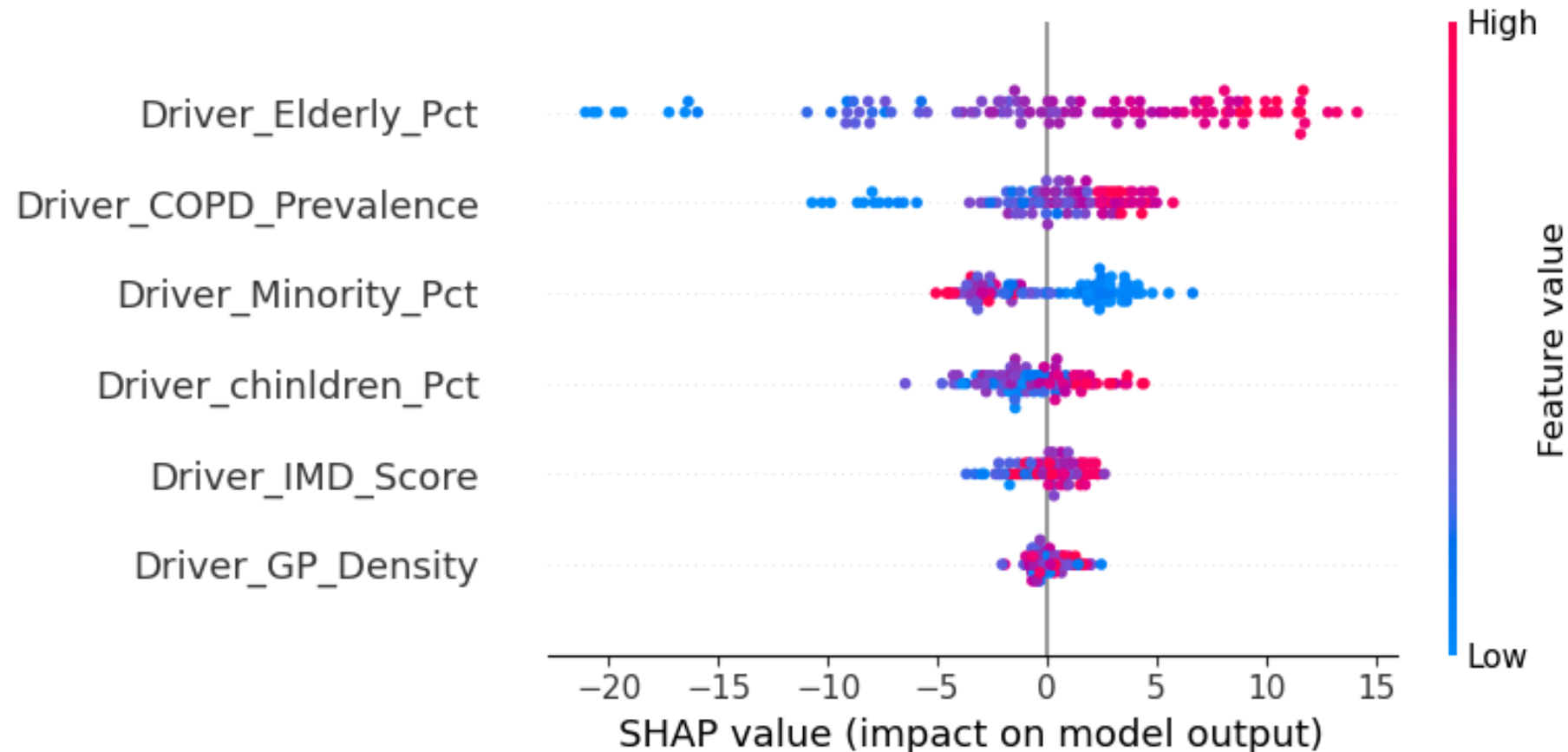
When all 106 regions are analyzed together, the percentage of the population over 65 has the greatest impact on antibiotic use.



A high elderly population strongly increases antibiotic use, while other factors show complex relationship.

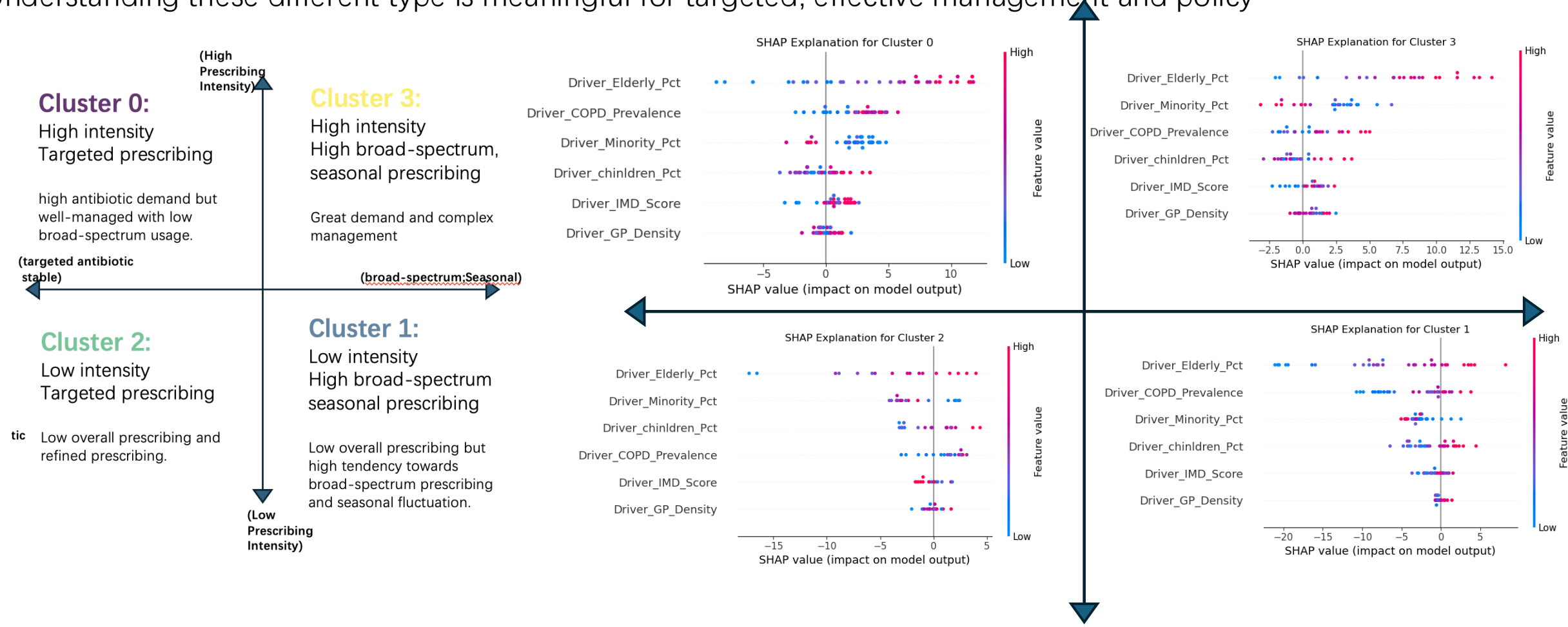
This plot shows the impact of each driver for each region.

For the elderly population, high values (red dot) strongly push antibiotic use higher (red side). In contrast, factors like ethnic minority percentage show a mixed impact at the national level.



But a single national profile is oversimplifying the situations.

Previous clustering analysis reveals that UK has 4 distinct clusters.
Each cluster with a unique combination of drivers for antibiotic use.
Understanding these different type is meaningful for targeted, effective management and policy



Cluster 3:

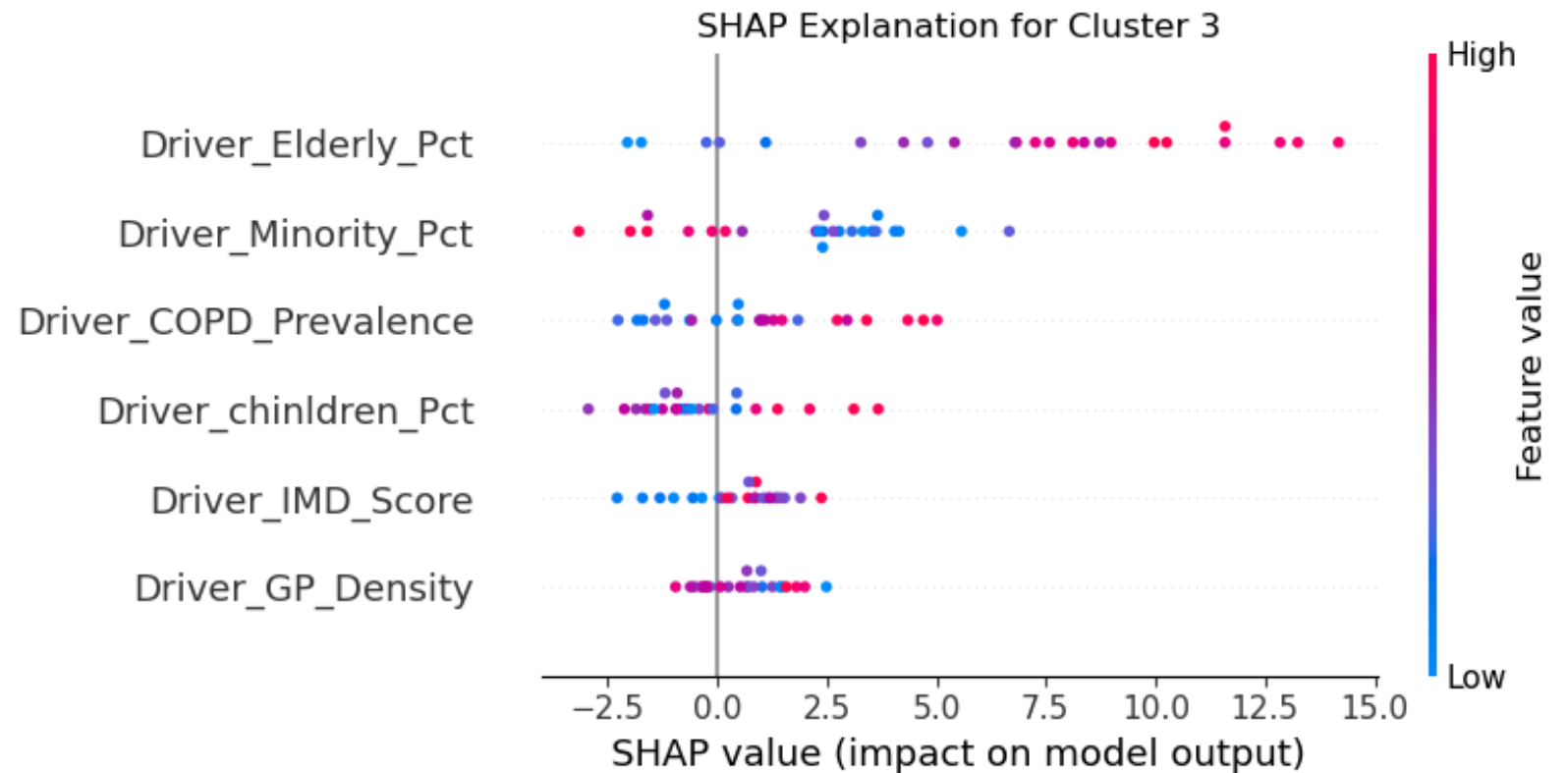
High intensity
High broad-spectrum,
seasonal prescribing

Great demand and complex
management

(broad-spectrum;Seasonal)

Double burden region: aging population, social complexity, GP shortage is factor in high antibiotic usage.

Aging population still is key driver for antibiotic use of cluster 3.
While minority percentage is somehow negatively interactive with use of antibiotic.
Lower GP density is associated with higher use antibiotic which is significant in this cluster.

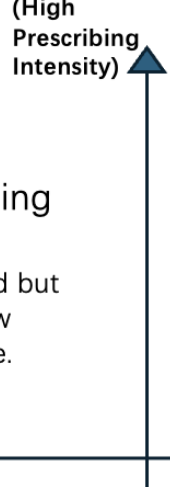


Cluster 0:

High intensity
Targeted prescribing

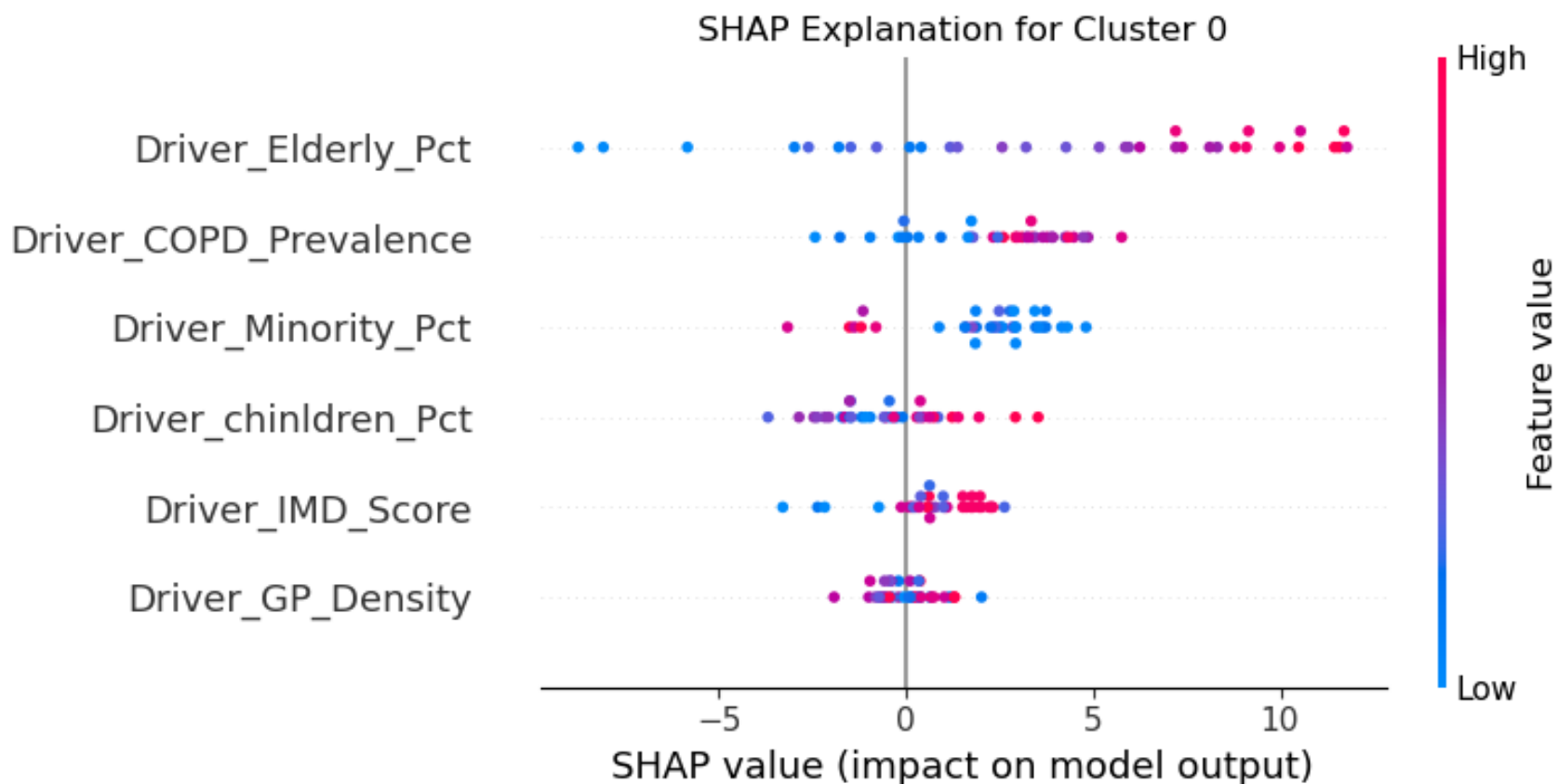
high antibiotic demand but
well-managed with low
broad-spectrum usage.

(targeted antibiotic
stable)



High demand regions : driven by aging population.

The impact of the elderly
population over all other factors.
COPD prevalence is a secondary,
but still significant, positive driver.



Cluster 2:

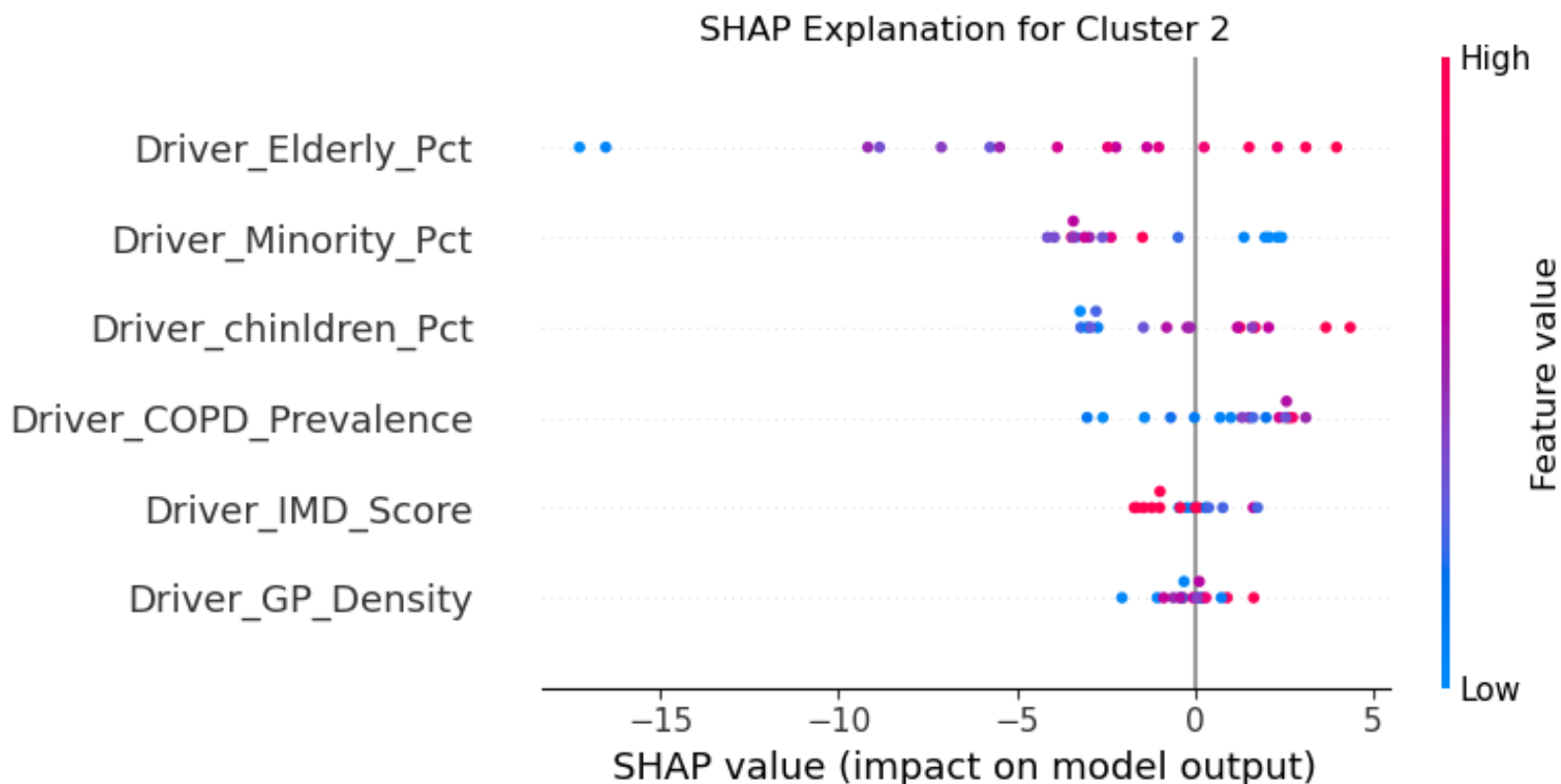
Low intensity

Targeted prescribing

Low overall prescribing and refined prescribing.

Safe regions: but with some hidden risk under low consumption Paradox.

In these regions, their peaceful mode is primarily due to lower elderly population. However, abnormally, high IMD score is associated with less antibiotic use, which leads concerns about accessibility of antibiotics.



(broad-spectrum;Seasonal)

Cluster 1:

Low intensity
High broad-spectrum
seasonal prescribing

Low overall prescribing but
high tendency towards
broad-spectrum prescribing
and seasonal fluctuation.

Young but inefficient regions

Low antibiotic consumption is
credited to young population. But
they don't have good manner for
prescribing appropriate antibiotic.

